

Construct Validity for Geometric Perturbation Scores: When Does a Number License a Mechanistic Claim?

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Abstract

Background: Geometric perturbation scores—such as direction instability, which measures the angular consistency of perturbation signatures across biological contexts—are used to rank drugs and genes by mechanism conservation. These scores predict biological properties, but what a given score *licenses* as a mechanistic claim depends on validity conditions that the raw number does not encode. We propose a construct-validity framework (a “validity ladder”) with seven conditions, of which three are tested empirically, that specifies which interpretive threats a score has survived.

Methods: We conducted pre-registered experiments across three perturbation domains: 8,949 LINCS L1000 drug compounds, 1,676 Perturb-seq CRISPRi gene knockdowns, and 25,254 JUMP-CP Cell Painting morphological profiles. Five drug hypotheses and three genetic hypotheses were tested with Holm–Bonferroni correction within each domain family. Validity conditions included phenotype projection onto genetic perturbation directions, transport-stable variance correction with leave-one-out cross-validation, and essential-gene confound correction on Grassmannian subspaces.

Results: Domain-specific validity conditions preserved global rankings in all three domains (LINCS: $\rho > 0.99$; Perturb-seq: $\rho = 0.91$; JUMP-CP: $\rho = 0.99$) while upgrading inferential warrant at the item level. Phenotype-projected scores correlated with on-target genetic perturbation connectivity ($\rho = 0.38$, $p < 10^{-27}$) while raw scores did not ($\rho = -0.04$). Transport-stable scores outpredicted raw scores for held-out cell-line effects in all 66 valid leave-one-out folds (Wilcoxon $p = 8.2 \times 10^{-13}$; mean $\Delta\rho = +1.10$, 95% CI [1.00, 1.19]). Essential-gene correction revealed that universal-essential knockdowns *diverge* rather than converge across cell types ($d = 0.60$, $p_{\text{perm}} < 10^{-4}$), inverting the predicted confound. Of 5 drug hypotheses, 2 survived multiplicity correction, 1 was refuted, and 2 were informative nulls.

Conclusions: Validity conditions do not change geometric perturbation scores; they change what a practitioner can conclude from them. The construct-validity framework tested here—applicable to any geometric score that measures perturbation-response consistency across contexts—provides a structured procedure for accumulating interpretive evidence, separating scores that license mechanistic claims from scores that merely rank.

Keywords: construct validity, direction instability, geometric perturbation scores, LINCS L1000, Perturb-seq, Cell Painting, validity ladder, pre-registration, Fréchet variance, Grassmannian geodesic

1 Background

Geometric perturbation scores—metrics that quantify how a biological system’s response to a perturbation varies across experimental contexts—are used throughout computational biology to rank

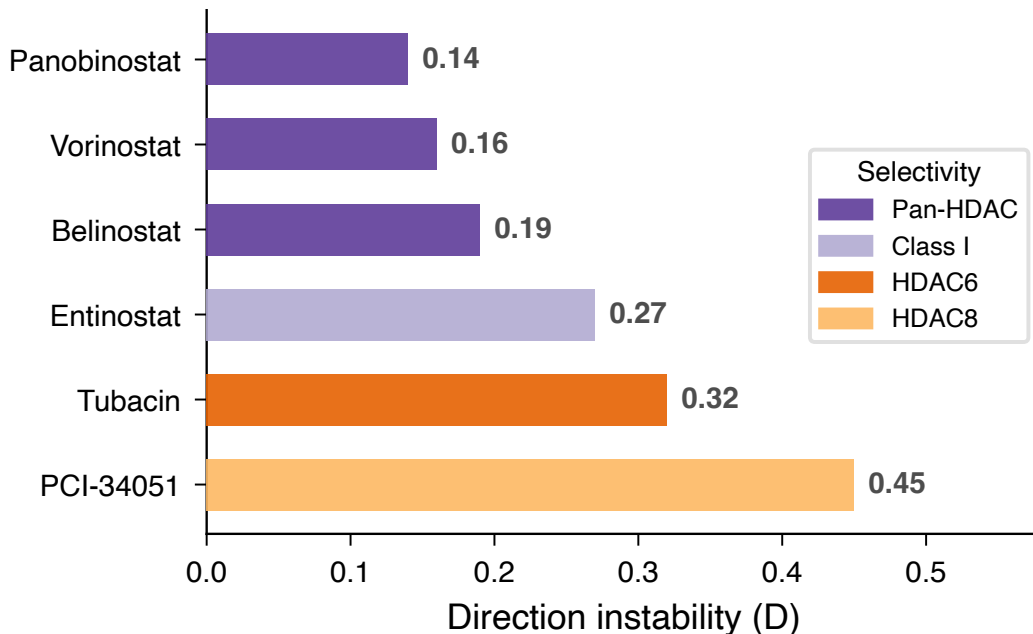


Figure 1: **HDAC selectivity gradient: calibration standard for direction instability.** Pan-HDAC inhibitors [5] (panobinostat, vorinostat, belinostat) achieve the lowest direction instability ($D = 0.14$ – 0.19); Class I-selective entinostat is intermediate ($D = 0.27$); isoform-selective HDAC6 and HDAC8 inhibitors (tubacin, PCI-34051 [6]) have the highest instability within the class ($D = 0.32$ – 0.45). This gradient, reproduced from Tower [3], provides a fine-grained rank ordering within the HDAC class: Spearman $\rho = 0.83$ between direction instability and selectivity rank across these 6 compounds after toxicity-gene correction. The HP1 “doesn’t bite” threshold ($\rho \geq 0.83$) was calibrated to this value.

drugs, genes, and compounds by mechanism conservation [1, 2]. Direction instability, one such score, measures the angular consistency of perturbation signatures across cell lines: a drug whose transcriptomic response points in the same direction regardless of cell type receives a low score, indicating conserved mechanism [3]. Applied to 8,949 drugs in LINCS L1000, direction instability produces biologically meaningful rankings: pan-HDAC inhibitors achieve the lowest scores ($D = 0.14$ – 0.19), with a selectivity gradient within the HDAC class (Figure 1) that tracks isoform specificity [4].

These rankings establish that the score captures genuine biological signal. They leave open a different question: under what conditions does a geometric perturbation score license a mechanistic *claim*? A score of $D = 0.55$ means signatures point in similar directions across cell lines. It does not mean the drug acts through a specific pathway or that the observation will replicate in a new context.

This gap between score and claim is a construct-validity problem [7, 8]. In Messick’s framework, a score’s meaning is not inherent in the number; it depends on accumulating evidence from multiple sources that the score measures what it purports to measure and that interpretations drawn from it are warranted. The same principle applies to geometric perturbation scores: a raw ranking may be reproducible and biologically correlated, yet insufficient to license mechanistic inference without additional validation evidence.

Three empirical threats illustrate this gap. First, a drug’s low direction instability could reflect

a coherent off-target program rather than on-target mechanism conservation—the raw score cannot distinguish the two. Second, a drug with low direction instability in training data may fail to predict a held-out cell line’s response, because the raw metric conflates genuine directional consistency with Fréchet variance from a small number of contexts. Third, in genetic perturbation screens, universal-essential knockdowns could inflate or deflate cross-cell-type distances through a shared lethal-response subspace, contaminating transport scores that appear mechanistically informative.

We formalize these threats through a validity ladder (Figure 3): a sequence of seven conditions that progressively strengthen what a geometric perturbation score licenses. Each rung answers a specific threat to interpretation. The ladder is a domain-specific instantiation of construct validation, analogous to the Bradford Hill criteria for causal inference [9, 10]: satisfying more rungs does not change the number, it changes what a practitioner is warranted in concluding from it. The rung ordering reflects logical dependency: a reliable measurement (Rung 2) is prerequisite for meaningful projection (Rung 5), and localized signal (Rung 4) precedes phenotype alignment (Rung 5). This ordering captures logical prerequisites rather than a strict hierarchy of evidence strength.

This paper tests whether validity conditions matter empirically across three perturbation domains: 8,949 LINCS L1000 drug compounds [2], 1,676 Perturb-seq genetic knockdowns [11], and 25,254 JUMP-CP Cell Painting morphological profiles [12]. Direction instability serves as the case study, but the construct-validity procedure—define threats, specify corrections, pre-register predictions, test whether corrections reorganize rankings or preserve them—applies to any geometric score measuring perturbation-response consistency across contexts. We report honest nulls alongside confirmations, because the pattern of where validity conditions matter—and where they do not—is itself the finding.

2 Definitions and notation

2.1 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

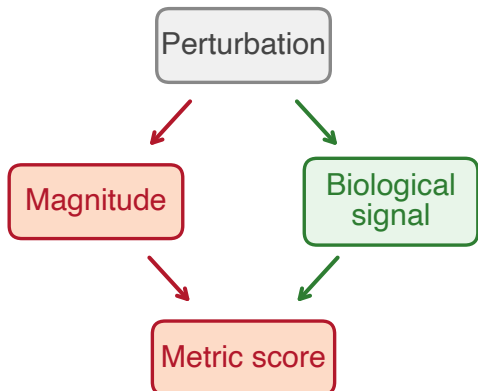
$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear. $D = 1$ when orthogonal on average. Cosine normalization removes the linear dependence on perturbation magnitude (Figure 2), under the assumption that signature direction and magnitude are separable—that is, that the direction a drug perturbs gene expression does not itself change with dose. Direction instability measures only direction [3]. Variance-based and estimation-based metrics conflate perturbation strength with biological signal. The separability assumption is tested empirically in the Discussion.

2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type (k = subspace dimension, d = ambient dimension). The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$, where $\boldsymbol{\theta}$ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, the pairwise average in Eq. 1 reduces to a single comparison, and the estimand carries no Fréchet variance or holonomy

a Confounded metrics



b Direction instability

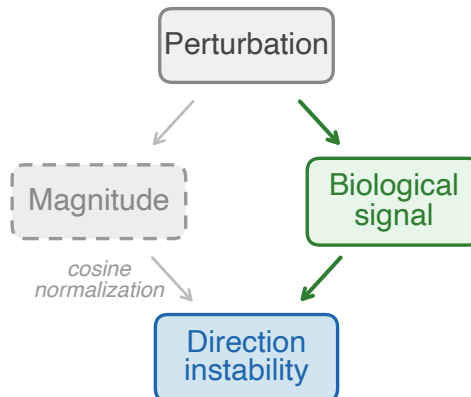


Figure 2: **Direction instability removes linear magnitude dependence.** (a) Confounded metrics (variance, subspace size) inherit effect magnitude through a shared path, producing spurious associations between perturbation strength and metric value. (b) Direction instability operates after cosine normalization, which blocks the linear magnitude path under the assumption that signature direction is independent of dose (separability). The observed association between group variable and metric reflects biological signal to the extent that this assumption holds. An empirical test of separability (dose-direction stability across 1,445 drugs) is reported in the Discussion.

information. We therefore do not compute direction instability as defined in Eq. 1 for Perturb-seq; we compute the Grassmannian geodesic distance $\delta(U_1, U_2)$ between per-cell-type response subspaces. This is a related but distinct geometric estimand, and we label it as such throughout rather than presenting it as direction instability at $K = 2$. Perturb-seq results are interpreted as a subspace-rotation analysis, not a direct replication of the LINCS vector-cosine analysis.

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting [12, 13] measures 3,180 morphological features per well across five fluorescent channels. JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A subset of 743 features—cell area, compactness, eccentricity, DNA/Mito intensity, granularity, and radial distribution—reflect general cell health rather than compound-specific morphological effects. The cell-health correction removes these features before recomputing direction instability.

2.4 The validity problem

Consider two drugs with identical direction instability, $D \approx 0.56$ (mechanism-conserving regime):

- **Drug A** is a topoisomerase II inhibitor. Its conserved direction aligns with its target gene’s knockdown signature—phenotype-projected bracket is high. The consistency comes from on-target DNA-damage signaling.

- **Drug B** is a proteasome inhibitor. Its conserved direction does *not* align with its annotated target subunit’s knockdown signature—phenotype-projected bracket is near zero. The consistency comes from broad proteostasis disruption shared across cell types.

Both achieve $D \approx 0.56$. Both pass the raw metric equally. Phenotype projection (Rung 5 of the validity ladder) distinguishes them: Drug A’s consistency is mechanistically on-target; Drug B’s is not. The distinction matters for downstream decisions: Drug A is a strong candidate for mechanism-based follow-up; Drug B requires additional investigation to identify what program drives its consistency before any mechanistic claim is warranted. A concrete instance of this scenario (daunorubicin vs. MG-132) appears in the Discussion.

2.5 The validity ladder

The validity ladder is a domain-specific instantiation of Messick’s construct-validation framework [8], adapted for geometric perturbation scores. We define seven rungs (Figure 3), ordered by logical dependency rather than by evidence strength:

1. **Process view declared.** What constitutes a “context”?
2. **Reliable vector field.** Adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** Effect isolated from background.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes.
5. **Phenotype-aligned.** Conservation direction matches a known target pathway.
6. **Transports across contexts.** Score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** Metric predicts an external outcome in new data.

Rungs 1–3 are satisfied by design in our datasets and are not tested further. Rungs 4–6 are the validity conditions we test empirically. Rung 7 (holdout prediction in a genuinely new dataset) is not tested: Experiment 5’s leave-one-cell-line-out cross-validation tests Rung 6 (transport within LINCS), not generalization to an independent dataset. A Rung 7 test would require predicting compound activity in a dataset collected independently of LINCS—such as a proprietary pharma screen or a future LINCS phase—using only validity-ladder scores computed from the current data.

3 Related work

Construct validity in measurement. Cronbach and Meehl introduced construct validity as a framework for evaluating whether a test measures its intended construct [7]. Messick’s unified framework [8] extended this into six facets—content, substantive, structural, generalizability, external, and consequential—establishing that a score’s meaning depends on accumulating evidence across multiple sources. Our validity ladder instantiates this framework for geometric perturbation scores: Rungs 1–3 address content and structural validity, Rungs 4–5 address substantive and external validity, and Rung 6 addresses generalizability. Recent construct-validity frameworks for computational scores [10, 14] and the SBB principles for perturbation-prediction models [15] share this orientation. Table 1 makes this mapping explicit.

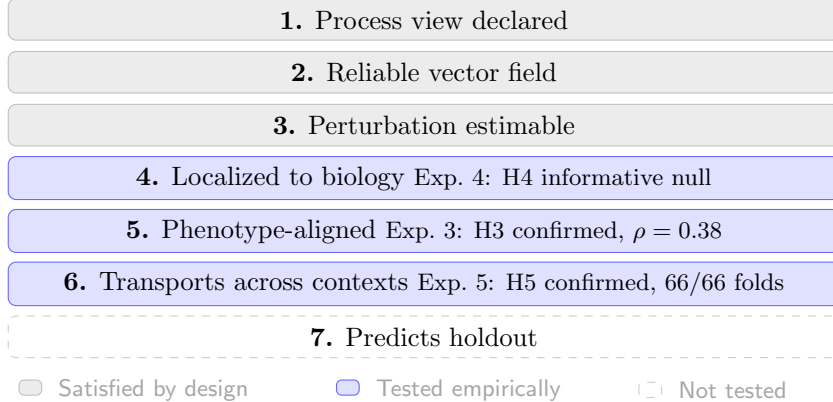


Figure 3: **The validity ladder.** Seven rungs progress from measurement prerequisites (1–3) through empirically tested validity conditions (4–6) to external prediction (7). Each rung eliminates a specific interpretive threat. Rungs 4–6 are tested across three domains (LINCS L1000, Perturb-seq, JUMP-CP); result summaries are shown at right. Satisfying additional rungs does not change the direction-instability score—it upgrades the inferential warrant attached to it.

Table 1: Mapping between Messick’s construct-validity facets and the validity ladder. The *consequential* facet is addressed qualitatively in the Discussion (warrant upgrade) rather than tested empirically. Rung 7 (external prediction in a new dataset) remains untested.

Messick facet	Ladder rung	What it licenses here
Content	1 — Process view declared	Context set is defined and appropriate
Substantive	2–3 — Reliable, estimable	Signal exceeds noise; effect is isolable
Structural	4 — Localized to biology	Score structure reflects pathway-relevant genes
External	5 — Phenotype-aligned	Score aligns with independent target signature
Generalizability	6 — Transports	Score replicates in held-out contexts
Consequential	(Discussion — warrant)	What a practitioner may act on

Perturbation biology. CERES [16] corrects copy-number artifacts in CRISPR screens; integrated Broad–Sanger analyses [17] document cross-study divergence. These corrections operate on dependency-score magnitudes; our contribution is orthogonal, correcting subspace geometry and showing that rank-preserving corrections carry inferential value. The Connectivity Map framework [1, 2] and subsequent metric evaluations [18] establish the transcriptomic-similarity scoring approach that direction instability extends. LEMUR [19] performs multivariate regression on Grassmann manifolds for multi-condition Perturb-seq; our Experiment 7 uses a simpler Grassmannian geodesic distance. Raju’s geometric coherence framework [20] applies related angular measures to single-cell CRISPR perturbations; our work differs in testing construct validity of such scores rather than using them directly for prediction. Pharmacogenomic interaction landscapes [21] and retrieval-based profile evaluation [22] provide complementary perspectives on cross-study consistency.

Metric evaluation in perturbation biology. Prior work has benchmarked connectivity and signature-similarity metrics for consistency and predictive value [2, 18, 23], construct-validity frame-

works have been proposed for ML benchmarks and computational scores [10, 14]; and recent work has shown that simple linear baselines match deep-learning perturbation predictors [24], while neural optimal transport [25] offers an alternative geometric approach to perturbation response modeling. These efforts evaluate *whether* a metric predicts a downstream label. Our contribution is orthogonal: we ask *what a given score licenses*—treating validity as a graded, condition-dependent warrant rather than a single predictive-accuracy number—and test where those conditions affect interpretation across three perturbation modalities. We are not aware of a directly comparable graded-warrant treatment applied to an unsupervised geometric perturbation score with pre-registered decision criteria.

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines), using Level 5 replicate-collapsed z-scores across 978 landmark genes [2, 26]. Of 77 cell lines in Phase I, approximately 50 contribute sufficient compound coverage for direction instability computation. We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines.

Perturb-seq data derive from Replogle et al. [11]: genome-scale CRISPRi knockdowns in K562 and RPE1 cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type (of these, 1,567 have matching DepMap Chronos labels for HP3; all 1,676 are used for HP1). Both screens target DepMap-essential genes; 85% of RPE1 targets overlap with K562 targets. Essentiality labels derive from DepMap Chronos scores [27]: 855 genes with Chronos < -0.5 in both K562 and RPE1 were classified as universal-essential; the remaining 712 are sub-threshold and serve as the comparison group. A label-permutation null (10,000 permutations) tested whether the observed group difference exceeds background K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery [12]: 25,254 compounds tested in ≥ 5 imaging sources after filtering (78% attrition from the 116,750 total; the majority of compounds appear in fewer than 5 sources, making direction instability—which requires multiple contexts—undefined for them). Cell-health features (743 features matching Area, Compactness, Eccentricity, DNA/Mito Intensity, Granularity, Radial Distribution, and Texture Entropy) were identified by substring matching on CellProfiler feature names; corrected direction instability was computed on the remaining 2,437 features.

Following pre-registration principles [28], all scoring functions and drug lists were committed before any real LINCS signatures were analyzed (initial freeze: commit 8d74bd0). Perturb-seq hypotheses and extended drug hypotheses (H5-full, H3-CRISPRi) were pre-registered before the correction analysis (commit f3c88ed, tag prereg-extended-psb, timestamped 2026-07-08T11:56:58–04:00, preceding experiment results committed at 14:42–04:00). The dose-direction separability test (Section 6.2) was pre-registered in commit 5bb6080 before execution. The repository was public at the time of all freezes.¹

Reporting guidelines. No existing reporting guideline (such as TRIPOD for clinical prediction models) directly governs a construct-validity study of an unsupervised geometric score. We

¹Code, pre-registration, deviation log, and all experimental scripts: <https://github.com/elliotttower/direction-instability-drug-validity>. LINCS data: GEO GSE92742; Perturb-seq: Replogle et al. [11]; JUMP-CP: Chandrasekaran et al. [12]; DepMap: Dempster et al. [27].

Table 2: Deviation log. All departures from the pre-registered plan, with rationale and timing. Full detail in `DEVIATION_LOG.md`, commit-linked.

#	Pre-registered	Actual	Why	When
1	Hypotheses frozen; gene-set method unspecified	Gene-set method declared in extended pre-reg	Original freeze omitted gene-set construction parameters for H3/H4/H5	Pre-analysis
2	H5: $\geq 4/5$ folds win	Threshold adjusts to $\lceil 0.8n \rceil$ if < 5 valid folds	Maintains 80% win rate when rare cell lines have too few drugs; not triggered in practice (71 cell lines)	Pre-analysis
3	H3 and H4 treated as independent	Shared shRNA-consensus dependency documented	Both derive from the same genetic perturbation object; not independent tests	Pre-analysis
4	Original H5: 5 alphabetical folds	Full 66-fold leave-one-out with Wilcoxon co-primary	Extended pre-reg (commit <code>f3c88ed</code>) replaces original	Pre-analysis
5	H3-CRISPRi: three-outcome decision tree	Outcome (c) realized ($\rho = -0.12$)	Followed frozen decision tree; K562-only confound explanation pre-registered	Post-analysis

adopt TRIPOD’s spirit where applicable—pre-specified analysis plan, frozen thresholds, complete reporting of all pre-registered hypotheses including nulls—and note deviations in Table 2. We omit items (such as patient-level calibration and outcome incidence) that do not apply to a non-clinical perturbation-screen setting.

We treat the five LINCS hypotheses (H1–H5) as one test family and the three Perturb-seq hypotheses (HP1–HP3) as a second. The separation reflects genuine independence: the two domains use different experimental systems (50+ cell lines vs. two specific lines), different perturbation types (small molecules vs. CRISPRi), different geometric estimands (vector cosines in $\mathbb{R}^{978}$ vs. Grassmannian geodesics in $\text{Gr}(5, 2073)$), and share no data or parameters. A single family across all eight hypotheses would be more conservative; we consider it overly so given the absence of any shared data or analytic dependence between domains. The choice is immaterial for the confirmed results: H3 ($p < 10^{-27}$) and H5 ($p = 8.2 \times 10^{-13}$) survive Holm–Bonferroni correction even under a single 8-hypothesis family. Experiment 8 (JUMP-CP) is reported separately because it uses a third data source with a distinct geometric estimand. After Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) and H5 ($p_{\text{Wilcoxon}} = 8.2 \times 10^{-13}$; also $p_{\text{perm}} < 10^{-3}$ on the preliminary 5-fold analysis) survive at $\alpha = 0.05$; H2 ($p = 0.28$) and H4 (gap = 0.018) do not; H1’s criterion-pass is an artifact. Within Perturb-seq, HP3 ($p_{\text{perm}} < 10^{-4}$) survives.

4.2 Phenotype-projected bracket (Experiment 3)

For each drug with an annotated target gene (Drug Repurposing Hub [29], 1,551 drugs), the “on-target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000 (4,369 genes eligible, ≥ 3 hairpins each, consensus averaged across up to 20 cell lines). Phenotype-

projected bracket projects the instability vector onto this on-target direction. On-target enrichment is defined as $\cos^2(\bar{\mathbf{s}}_{\text{drug}}, \bar{\mathbf{s}}_{\text{shRNA}})$: the squared cosine between each drug’s mean signature and its target gene’s shRNA consensus, measuring how much of the drug’s transcriptomic effect aligns with the genetic perturbation of its annotated target.

shRNA signatures have well-documented off-target effects [30], which add noise to the on-target direction and attenuate the observed correlation: the true on-target association is likely stronger than the measured $\rho = 0.38$. The shRNA consensus averages across up to 20 cell lines, producing a cell-type-agnostic on-target direction at the cost of off-target contamination. Similarly, Drug Repurposing Hub target annotations carry noise from multi-target compounds and legacy assignments; misannotation further attenuates the correlation. Polypharmacology compounds this: drugs with strong secondary targets may show phenotype projection onto the primary target even when cross-cell-line consistency is driven by a secondary mechanism, a confound that single-target annotations cannot resolve. A convergent-validity check using CRISPRi signatures (single cell type, cleaner loss-of-function but cell-type-specific) is reported in the Results.

The pre-registered prediction (H3) required Spearman $|\rho| > 0.3$ between phenotype-projected bracket and on-target enrichment, and $|\rho| < 0.15$ for raw bracket versus the same enrichment.

4.3 Transport-stable bracket (Experiment 5)

Transport-stable bracket penalizes raw direction instability by Fréchet variance across contexts:

$$\text{TS} = D - \lambda \cdot \text{Var}_{\text{Fréchet}} \quad (2)$$

where $\text{Var}_{\text{Fréchet}} = \frac{1}{K} \sum_i \arccos^2(\hat{\mathbf{s}}_i^\top \bar{\mathbf{s}})$ and $\bar{\mathbf{s}}$ is the Fréchet mean direction. The subtraction is a heuristic combination: the alternative (a ratio) would be undefined when $\text{Var}_{\text{Fréchet}} = 0$ and would amplify noise for highly consistent drugs. The penalty weight λ was fixed at the value from Tower [3] and was not optimized for this evaluation.

We evaluate via leave-one-cell-line-out cross-validation over all cell lines with ≥ 20 eligible drugs after excluding the held-out cell line (66 valid folds out of 71 total; 5 excluded for insufficient drug count). A preliminary 5-fold analysis using alphabetically-selected cell lines (A375, A549, A673, AGS, ASC) established the rare/common regime distinction with permutation nulls; the full leave-one-out extends this to all cell lines and adds Wilcoxon signed-rank as a co-primary criterion.

The pre-registered prediction (H5) required TS to outpredict raw bracket in $\geq 80\%$ of valid folds (co-primary 1), with Wilcoxon signed-rank on per-fold $\Delta\rho$ rejecting H_0 : median ≤ 0 at $\alpha = 0.05$ (co-primary 2).

4.4 Essential-gene correction on Perturb-seq (Experiment 7)

We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in K562 and RPE1) via iterative projection on $\text{Gr}(5, 2073)$ (100 iterations, convergence to $\Delta < 10^{-6}$ in Grassmannian distance; the result is insensitive to initialization because the essential-gene subspaces cluster tightly). We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances.

Two pre-registered predictions were tested. HP1 required Spearman ρ between raw and corrected distances below 0.70 (ranking-disruptive, denoted “bites”) or ≥ 0.83 (ranking-preserving, denoted “doesn’t bite”). We use these shorthand terms throughout: a correction “bites” when it reorganizes the population ranking ($\rho < 0.70$) and “doesn’t bite” when it preserves it ($\rho \geq 0.83$). The ranking-preserving threshold of 0.83 was calibrated to the HDAC selectivity gradient (Figure 1): the finest within-class rank ordering that direction instability resolves. The ranking-disruptive

threshold of 0.70 was set as a round-number boundary below which more than half the variance is unexplained ($R^2 < 0.49$), following the companion paper [3]. Values between 0.70 and 0.83 would require further analysis to interpret. HP3 predicted that universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$).

4.5 Cell-health correction on JUMP-CP (Experiment 8)

The 743 cell-health features constitute 23% of the 3,180-feature space—proportionally six times larger than the LINCS toxicity gene set (41/978, 4%) and the largest feature removal in this study. If cell-health variance dominates cross-source consistency, this correction should reorganize the ranking more substantially than the drug or genetic corrections. This makes Experiment 8 the strongest test of the thesis: if corrections don’t bite even when the removed feature set is proportionally large, the ranking is robust.

The pre-registered prediction used the same threshold: Spearman ρ between raw and corrected direction instability below 0.70 (“bites”) or ≥ 0.83 (“doesn’t bite”).

4.6 Supplementary experiments

Three additional drug-domain experiments are reported in Additional file 1. **Experiment 1** (H1, toxicity correction) tests whether cytotoxic drugs masquerade as mechanism-conserving. **Experiment 2** (H2, broad-mechanism drugs) tests whether broad-target therapeutics fall below the population median. **Experiment 4** (H4, localization score) tests whether gene-set restriction improves MOA prediction. HP2 (GO pathway prediction from corrected Perturb-seq distances) is also in Additional file 1.

5 Results

5.1 Pre-registered scorecard

Table 3 summarizes all pre-registered hypotheses. Two of five drug hypotheses survive Holm–Bonferroni correction (H3, H5); H2 meets its pre-registered criterion but does not survive multiplicity correction; H1 is refuted with an artifact criterion-pass; H4 is an informative null. In the Perturb-seq domain, HP3 is confirmed; HP1 shows the correction “doesn’t bite” at the population level; HP2 is a null. Experiment 8 (JUMP-CP) shows the correction “doesn’t bite.”

5.2 Experiment 3: Phenotype-projected bracket separates on-target from off-target instability

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected bracket correlates with on-target enrichment at Spearman $\rho = 0.376$ ($p = 4.8 \times 10^{-28}$). Raw bracket shows no correlation ($\rho = -0.043$, $p = 0.22$). Both pre-registered criteria are satisfied.

H3 confirms Rung 5 of the validity ladder. Raw direction instability measures whether a drug’s signatures point in consistent directions, but this consistency could arise from any coherent program. Projecting onto the genetic perturbation direction of the annotated target isolates the mechanistically relevant component. The near-zero raw correlation shows that global instability carries no on-target information; the strong projected correlation shows that the on-target projection recovers it.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus is an empirical object from the same LINCS platform, requiring no pathway database selection. H3

Table 3: Pre-registered scorecard. *Criterion* = did the result meet the frozen pre-registered threshold? *Holm* = does it survive Holm–Bonferroni correction within its domain family ($\alpha = 0.05$)? *Warrant* = the interpretive status after both checks. The three columns earn their keep because H1 and H2 dissociate: both meet their pre-registered criteria, yet H1 is refuted (criterion-pass is an artifact) and H2 does not survive multiplicity correction. Result details for starred entries appear in Section 5.6.

	Prediction	Result	Criterion	Holm	Warrant
H1*	Toxicity reclassifies	$\Delta < 0.01$	Met (artifact)	—	Refuted —artifact of pre-existing high instability
H2*	Broad drugs below median	7/10 below	Met	No	Suggestive —does not survive multiplicity
H3	$ \rho_{\text{proj}} > 0.3$	$\rho = 0.376$	Met	Yes	Confirmed
H4*	AUROC gap > 0.05	Gap = 0.018	Not met	No	Informative null —domain-dependent
H5	TS wins $\geq 80\%$ folds	66/66, $p_W = 8.2 \times 10^{-13}$	Met	Yes	Confirmed
HP1	Correction bites ($\rho < 0.70$)	$\rho = 0.91$	Not met	—	Doesn’t bite
HP2*	GO prediction improves	$\Delta\rho = -0.02$	Not met	—	Doesn’t bite
HP3	Essential $d > 0$	$d = 0.60$	Met	Yes	Confirmed (inverted confound)
Exp. 8	Health correction bites	$\rho = 0.987$	Not met	—	Doesn’t bite

and H4 share this shRNA dependency (documented in DEVIATION_LOG.md entry 3) and are not independent tests. The phenotype-projected bracket is applicable only to drugs with target annotations—1,551 of 8,949 (17%). For the remaining 82% of drugs with unknown targets, the validity ladder provides other rungs (transport stability, localization) that do not require target annotations.

As a pre-registered convergent-validity check, we recomputed phenotype-projected bracket using CRISPRi loss-of-function signatures [11] as the on-target direction. The projected-bracket correlation with on-target enrichment did not reach significance (Spearman $\rho = -0.12$, $p = 0.16$, $n = 131$), corresponding to pre-registered outcome (c). On the 114-drug matched subset (drugs with both ground truths), shRNA projection yields $\rho = 0.42$ while CRISPRi projection yields $\rho = -0.14$, confirming that the ground-truth source—not the drug set—drives the difference.

The pre-registered secondary criterion points to the likely cause: raw bracket correlated positively with CRISPRi enrichment ($\rho = +0.27$, $p = 0.002$) while projected bracket did not—the reverse of the shRNA pattern. This reversal is consistent with the pre-registered K562-only confound: CRISPRi signatures derive from a single hematopoietic cell type, so the CRISPRi “on-target direction” captures cell-type-specific effects rather than cross-context mechanism—a confound that multi-cell-line shRNA consensus avoids by averaging across ~ 20 cell lines. One plausible explanation for the positive raw-bracket/CRISPRi correlation, which we did not test directly, is that K562 expresses many drug targets at high levels, producing spurious alignment between raw instability and K562-specific signatures.

CRISPRi does not provide independent convergent validity for H3. The shRNA-based result stands on its original evidence rather than being reinforced. An alternative concern is that the

Table 4: Leave-one-cell-line-out cross-validation (selected folds from the full 66-fold analysis). Transport-stable (TS) bracket outpredicts raw bracket in all folds. In common folds (8,000+ drugs), both metrics are anti-predictive but TS is less so. In rarer folds, TS reverses the sign entirely. Full 66-fold results: mean $\Delta\rho = +1.10$, Wilcoxon $p = 8.2 \times 10^{-13}$.

Held-out cell	N_{drugs}	ρ_{raw}	ρ_{TS}	Winner
A375	8,328	-0.480	-0.155	TS
A549	8,888	-0.494	-0.201	TS
A673	357	-0.725	+0.725	TS
AGS	356	-0.712	+0.714	TS
ASC	2,345	-0.365	+0.346	TS

H3 correlation is partly tautological: both drug signatures and shRNA ground-truth signatures derive from the same LINCSeq platform and largely overlapping cell lines, so shared platform-specific variance could inflate the association. The near-zero *raw*-bracket correlation ($\rho = -0.04$) argues against this: if platform sharing drove the result, raw bracket should also correlate with shRNA enrichment, since it uses the same signatures. The correlation emerges only after phenotype projection, suggesting it reflects on-target alignment rather than shared platform noise.

HDAC-removal sensitivity. Pan-HDAC inhibitors are the calibration standard for direction instability (Figure 1), so the H3 correlation could be driven by this single drug class. To test this, we removed all 20 HDAC-targeting drugs (spanning HDAC1, HDAC2, and HDAC6 targets; full list in Additional file 1) and recomputed the phenotype-projected correlation. The result is essentially unchanged: $\rho = 0.3759$ ($n = 775$, $p = 2.0 \times 10^{-27}$), compared with $\rho = 0.3756$ on the full 795-drug set ($\Delta\rho = 3 \times 10^{-4}$). A synonym check for “histone deacetylase,” “class I HDAC,” “pan-HDAC,” and “sirtuin” confirmed no drugs were missed by the primary filter.

Removing the 20 drugs with lowest raw bracket (most directionally consistent regardless of class) also preserves the correlation ($\rho = 0.387$, $p = 4.8 \times 10^{-29}$). Removing the 20 drugs with highest projected bracket—a direct test of whether a handful of phenotype-aligned outliers carry the signal—yields $\rho = 0.350$ ($p = 9.8 \times 10^{-24}$). All three removal analyses exceed the pre-registered threshold of $\rho > 0.20$. Bootstrap 95% CI on the full set is [0.315, 0.433], excluding zero. The H3 correlation is broad-based, not driven by HDAC inhibitors or a small number of extreme drugs.

5.3 Experiment 5: Transport-stable bracket outpredicts raw bracket in held-out cell lines

Transport-stable bracket outpredicts raw bracket in all 66 valid leave-one-out folds. Both co-primary criteria are satisfied: TS wins 66/66 folds (100%, criterion $\geq 80\%$) and Wilcoxon signed-rank on per-fold $\Delta\rho$ yields $p = 8.2 \times 10^{-13}$ (criterion $p < 0.05$).

Across 66 folds, mean raw $\rho = -0.60$ (anti-predictive in every fold; range $[-0.78, -0.22]$) while mean TS $\rho = +0.50$ (positively predictive). Mean $\Delta\rho = +1.10$ (95% bootstrap CI [1.00, 1.19]; all 66 values positive, range [0.17, 1.57]). Five cell lines were excluded for having < 20 eligible drugs. Figure 4 shows the $\Delta\rho$ distribution and the frequency gradient; Table 4 shows representative folds illustrating the rare/common regime distinction.

In rarer folds (such as A673, AGS, ASC), raw bracket is strongly anti-predictive ($\rho = -0.37$ to -0.73): drugs with high raw instability in training produce held-out signatures that point *away*

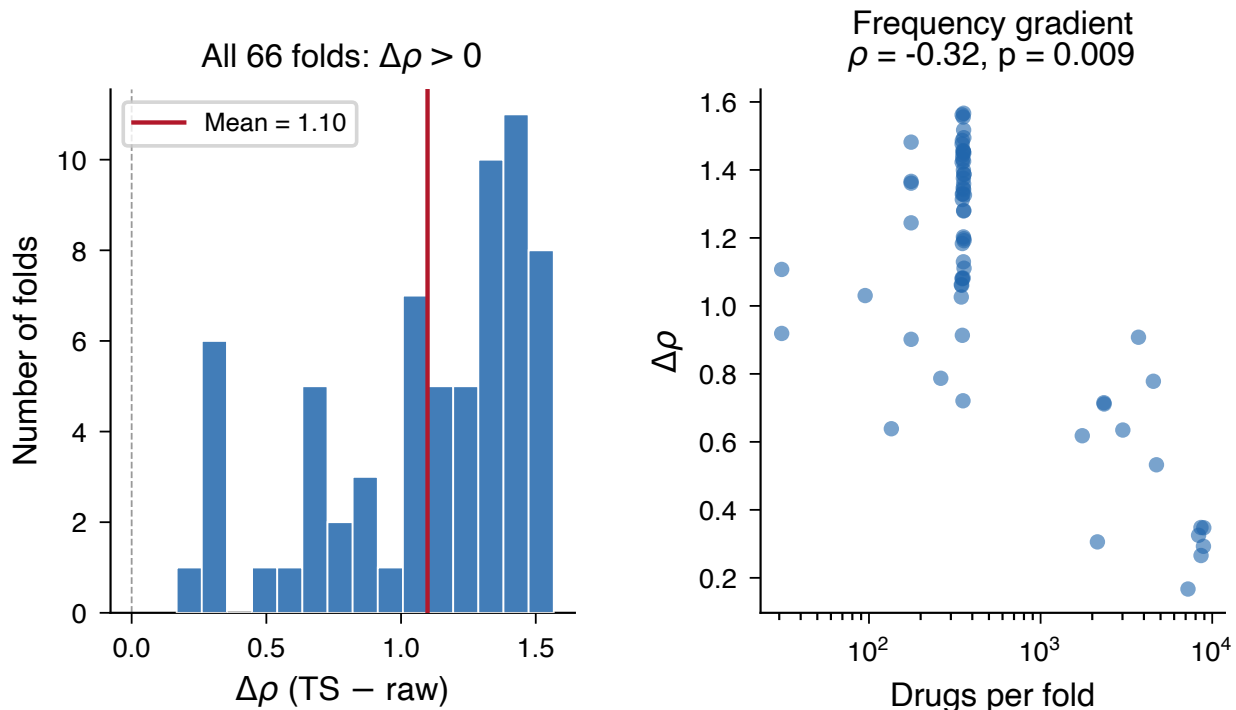


Figure 4: **Full 66-fold leave-one-out: TS advantage is universal.** **Left:** Distribution of per-fold $\Delta\rho$ (TS - raw Spearman correlation with held-out cosine). All 66 values are positive; the mean (red line) is +1.10. **Right:** Rarer cell lines (fewer drugs per fold) show larger TS advantages (Spearman $\rho = -0.32$, $p = 0.009$), confirming that the rare/common regime distinction from the preliminary 5-fold analysis generalizes across all cell lines.

from the training consensus. Transport-stable bracket reverses this, becoming positively predictive ($\rho = +0.35$ to $+0.73$). The reversal occurs because removing a single rare cell line substantially changes the consensus direction; drugs whose training instability was driven by that cell line are precisely those that TS downweights via the Fréchet penalty. This pattern holds across all 66 folds: the frequency-improvement correlation (Spearman $\rho = -0.32$, $p = 0.009$) confirms that rarer cell lines show larger TS advantages.

In common folds (8,000+ drugs), both metrics remain anti-predictive. TS improves over raw ($\rho = -0.155$ vs. -0.48 for A375; -0.201 vs. -0.49 for A549), but both correlations stay negative. TS “wins” in these folds by losing less badly, not by becoming positively predictive. The consensus direction is stable when computed from many cell lines, leaving less room for the Fréchet penalty to improve prediction.

Raw direction instability is anti-predictive because it conflates two sources of angular consistency: genuine mechanism conservation (signal) and the geometric tightness of the consensus direction around a small number of contexts (noise). Drugs with low raw DI in training tend to have tight consensus directions that are sensitive to removing any single cell line—precisely the drugs that fail to predict held-out effects. This is a failure of the raw estimator when applied to leave-one-out prediction without variance correction, not a failure of direction instability as a concept.

Before conducting the full 66-fold leave-one-out, we tested whether the TS advantage could be algebraic using a variance-preserving permutation null on the preliminary 5-fold analysis. Held-out

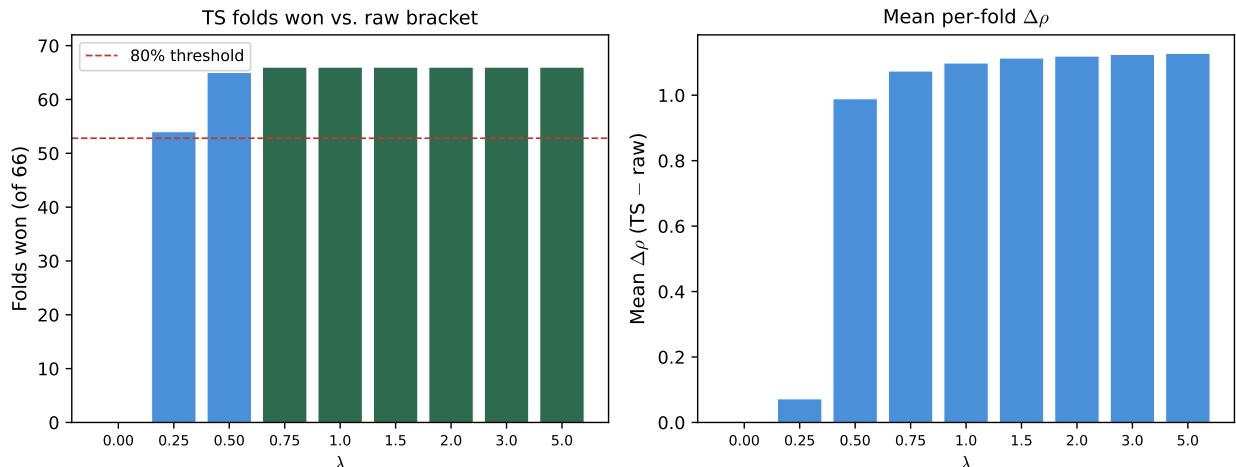


Figure 5: **Lambda sensitivity analysis for transport-stable bracket.** **Left:** Number of folds (out of 66) where TS outpredicts raw bracket, by λ . The dashed line marks the pre-registered 80% threshold (52.8 folds). At $\lambda = 0$ (sanity check), $\text{TS} \equiv \text{raw}$ and wins 0 folds. All eight non-zero λ values pass the criterion. **Right:** Mean per-fold $\Delta\rho$ ($\text{TS} - \text{raw}$) increases steeply from $\lambda = 0$ to $\lambda = 0.5$ and plateaus above $\lambda = 1.0$, indicating diminishing returns from stronger variance penalization.

signatures are shuffled across drugs while training data (raw DI, Fréchet variance, TS, consensus vector) remain fixed, and held-out cosines are recomputed against each drug’s original consensus. Under this null (1,000 permutations), TS wins only 0.1 of 5 folds on average with mean $\Delta\rho = -0.12$ (raw DI *outpredicts* TS when drug-specific signal is broken). The observed advantage falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$). The full 66-fold Wilcoxon test ($p = 8.2 \times 10^{-13}$) provides independent confirmation that the advantage is drug-specific.

Lambda sensitivity. The penalty weight $\lambda = 1.0$ was inherited from the companion paper and not optimized for this evaluation. To test whether the H5 result depends on this specific value, we repeated the full 66-fold cross-validation at $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 5.0\}$. At $\lambda = 0$, TS reduces to raw bracket exactly; the sanity check confirms 0/66 wins (strict inequality on identical vectors is deterministically false). At $\lambda = 0.25$, the smallest non-zero value, TS already wins 54/66 folds (82%, Wilcoxon $p = 9.5 \times 10^{-8}$). All eight non-zero λ values pass the pre-registered criterion ($\geq 80\%$ wins and Wilcoxon $p < 0.05$), far exceeding the “ ≥ 5 of 8” success threshold. For $\lambda \geq 0.75$, TS wins all 66 folds ($p = 8.2 \times 10^{-13}$). The 66/66 win rate saturates at a ceiling; the per-fold $\Delta\rho$ distribution provides finer resolution of the advantage magnitude across λ values. Figure 5 summarizes the sensitivity analysis. The H5 result is robust to penalty weight choice across a 20-fold range.

5.4 Experiment 7: Essential-gene correction preserves ranking but reveals confound-dominated core-machinery genes

Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 sub-threshold). Cohen’s $d = 0.60$ is a medium effect—the distributions overlap substantially, and

individual genes span the full range in both groups. The finding is that the *group means* differ in the opposite direction from the predicted confound, not that essentiality status determines geodesic distance.

Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1 (Figure 6). This is consistent with cell-line-specific essentiality effects documented in CERES [16] and integrated Broad–Sanger analyses [17]: the functional consequences of losing a core-machinery gene depend on the cell’s lineage and metabolic context.

A label-permutation null (shuffling essential/non-essential labels across the 1,567 genes, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval $[-0.021, 0.022]$, $p_{\text{perm}} < 10^{-4}$. The null preserves the geodesic-distance distribution while breaking only the label assignment, so the $\sim 6\times$ exceedance of the null half-width rules out the library-overlap confound (85% target overlap between screens).

Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-100}$). The global ranking is preserved.

The correction’s value is at the item level. The top 100 rank-shifting genes are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$). Proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 rank positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. The group-mean comparison (essential vs. non-essential, Cohen’s $d = 0.60$) provides supporting context: the distributions overlap substantially, and essentiality status is a weak per-gene classifier. HP3 is best understood as a shift in group-level geometry localized to identifiable core-machinery complexes, not a gene-level discriminator. These complexes are the genes whose cross-cell-type geometry is most dominated by the shared essential-response axis; the correction provides a concrete action: flag them before interpreting their transport scores.

5.5 Experiment 8: Cell-health correction has negligible effect in JUMP-CP

Among 25,254 compounds, corrected direction instability tracks raw at Spearman $\rho = 0.987$ ($p < 10^{-100}$). By the pre-registered criterion, this “doesn’t bite”—the strongest result for ranking robustness across all three domains.

The 743 health features constitute 23% of the 3,180-feature space—proportionally six times larger than the drug toxicity gene set (41/978, 4%)—yet produce *less* ranking impact ($\rho = 0.987$ vs. $\rho > 0.99$ for LINCS). That a proportionally larger feature removal barely moves the ranking is stronger evidence for the thesis than the drug case: cell-health features carry less of the cross-source variance in morphological signatures than stress genes carry in transcriptomic signatures. As noted in Section 4.1, the 78% attrition from the full JUMP-CP dataset (116,750 to 25,254 compounds) means the retained set is biased toward well-replicated compounds; the result may not generalize to sparsely-profiled compounds.

One explanation for the negligible effect: in morphological space, “health” and “mechanism” features are less separable than in transcriptomic space. A drug that alters cell shape changes both health-associated features (area, compactness) and mechanism-specific features (texture, radial distribution) simultaneously, because morphological effects are inherently coupled. In 978-gene transcriptomic space, stress-response genes and pathway-specific genes are more separable, which is why the LINCS toxicity correction has a larger (though still small) effect.

The near-perfect global correlation coexists with individual compounds shifting up to 7,155 rank positions, identifying those whose apparent morphological consistency is health-feature-dominated.

The cell-health feature definition (substring matching on CellProfiler names) is a limitation: a partition based on correlation with viability assays would be more principled and could yield different correction magnitudes.

5.6 Summary of supplementary experiments

Full methods, tables, and discussion appear in Additional file 1.

H1 (toxicity correction). Among 13 cytotoxic drugs, most are already high-instability ($D > 0.87$ for 9 of 13); toxicity correction shifts values by < 0.01 . The predicted failure mode—cytotoxicity masquerading as mechanism conservation—does not hold. Cytotoxicity produces genuinely context-dependent signatures in 978-gene space.

H2 (broad-mechanism drugs). Seven of 10 broad-mechanism drugs fall below the population median. Pan-HDAC inhibitors dominate ($D = 0.49$ – 0.55). The pre-registered second prediction—that kinase inhibitors would fall above the median—failed. The sorting axis is target breadth, not clinical specificity.

H4 (localization). Macro-averaged AUROC gap is $+0.018$ (below $+0.05$ threshold). Localization improves prediction for intracellular enzyme inhibitors (topoisomerase: $+0.49$, CDK: $+0.26$) but hurts for receptor-mediated drugs (acetylcholine: -0.20 , PPAR: -0.19). Rung 4 is domain-dependent.

HP2 (pathway function prediction). GO Jaccard overlap has near-zero correlation with geodesic distance ($|\rho| < 0.025$) for both raw and corrected forms. Geometric similarity of perturbation responses is distinct from functional pathway relatedness.

6 Discussion

6.1 Construct validity separates score from claim

The central finding is negative in the most useful sense: validity conditions do not reorganize the direction-instability ranking. This thesis was genuinely at risk: had any correction produced $\rho < 0.70$ while simultaneously improving prediction of an external outcome, the “reclassify” interpretation would have prevailed—the raw ranking would have been wrong, not merely underwarranted. That no correction crossed this threshold across three domains is the empirical result. The Perturb-seq result involves a degenerate $K = 2$ pairwise comparison rather than genuine $K \geq 5$ angular consistency; the qualitative conclusion holds across domains, but the genetic domain tests a weaker geometric estimand.

A rank-preserving correction (“doesn’t bite”) admits two interpretations: the targeted confound is absent, or the confound is present but rank-preserving by construction. These are distinguishable when item-level rank shifts are examined: if the correction produces large individual shifts (such as proteasome/spliceosome genes shifting 660–900 positions in Perturb-seq) while preserving the global correlation, the confound is real but concentrated in identifiable items. The population-level ρ and the item-level rank shifts together provide richer evidence than either alone.

What changes is the inferential warrant. Phenotype projection (H3) shows that direction instability decomposes into on-target and off-target components ($\rho = 0.38$ vs. -0.04); a CRISPRi convergent-validity check does not independently confirm this finding, likely because K562-only CRISPRi signatures are confounded by cell-type-specific effects. Transport stability (H5) shows the metric predicts held-out effects only after Fréchet variance correction; raw bracket is anti-predictive in all 66 folds. Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while the population ranking holds.

“Warrant upgrade” is a concrete claim about downstream decisions—Messick’s consequential validity facet—not a philosophical abstraction. The asymmetry is deliberate: failing a rung does not downgrade a drug’s ranking or remove it from consideration; it flags the drug for further investigation before a mechanistic claim is warranted. Consider two scenarios:

Drug repurposing triage. Daunorubicin (topoisomerase II inhibitor, 19 cell lines) and MG-132 (proteasome inhibitor, 16 cell lines) have nearly identical raw direction instability ($D = 0.557$ vs. 0.565 , $\Delta = 0.008$). Raw bracket ranks them equivalently—both in the mechanism-conserving regime. Phenotype-projected bracket separates them: daunorubicin scores 11.1 with on-target enrichment 0.24 (its consistency aligns with *TOP2A* knockdown); MG-132 scores 3.5 with enrichment < 0.001 (its consistency does not project onto *PSMB1* knockdown). One interpretation, consistent with proteasome-inhibitor pharmacology, is that MG-132 produces consistent transcriptomic effects through broad proteostasis disruption rather than through the specific program of its annotated target subunit. A practitioner using raw bracket alone would treat both drugs identically. After Rung 5 validation, daunorubicin is prioritized for mechanism-based follow-up; MG-132 requires investigation of what program drives its consistency. This illustrative pair was selected by matching raw bracket ($|\Delta D| < 0.05$) and ranking by projected-bracket divergence; individual drug-level scores carry uncertainty from cell-line sampling ($n = 16\text{--}19$).

Perturb-seq hit triage. Among 100 genes with low geodesic distance, 15 shift >500 rank positions after essential-gene correction (proteasome, spliceosome components). These are flagged as potentially confound-dominated. The remaining 85 genes have the same transport scores but stronger warrant—they are prioritized for functional validation. The warrant upgrade changes which genes proceed to follow-up.

6.2 What direction instability measures

Under the DAG of Figure 2, cosine normalization blocks the linear magnitude path from perturbation strength to the metric. Direction instability operates in a specific regime: it compares signatures *across cell lines at a fixed dose*. In this regime, magnitude variation arises from cell-line-specific pharmacodynamics (how strongly each cell line responds), and cosine normalization removes it. The separability assumption—that signature direction is independent of response magnitude—holds trivially when the dose is fixed, because the “magnitude” that varies across cell lines is response amplitude, not concentration.

A stronger version of separability would require direction to be independent of dose concentration. A pre-registered empirical test (commit 5bb6080) assessed this stronger claim across 6,169 drug $\times$ cell-line pairs profiled at ≥ 2 doses (1,445 drugs, 52,920 dose-pair comparisons). The median within-pair cosine between dose-level signatures was 0.19, and 86.5% of pairs fell below 0.5—both pre-registered criteria failed (thresholds: median > 0.7 , $< 20\%$ below 0.5).

These low cosines are dominated by signature-level noise rather than genuine dose-dependent direction changes. Of dose-level groups in this analysis, 80% contain a single replicate; in 978 dimensions, two single-replicate z-score vectors have expected cosine near zero even when the underlying true direction is identical. A diagnostic control confirms this: near-identical doses (ratio < 1.01 , such as 9.99 vs. 10.00 μM —the same intended dose with measurement imprecision) yield median cosine = 0.11, establishing the noise floor. The overall median of 0.19 is barely above it. The pre-registered criteria cannot be met at this replicate depth, and the test lacks statistical power to assess separability.

The separability assumption therefore remains empirically unresolved: the test was underpowered for single-replicate data in 978 dimensions. A properly powered test would require ≥ 3 replicates per dose level and a within-dose replicate baseline—conditions that the LINCS Level 5 archive does

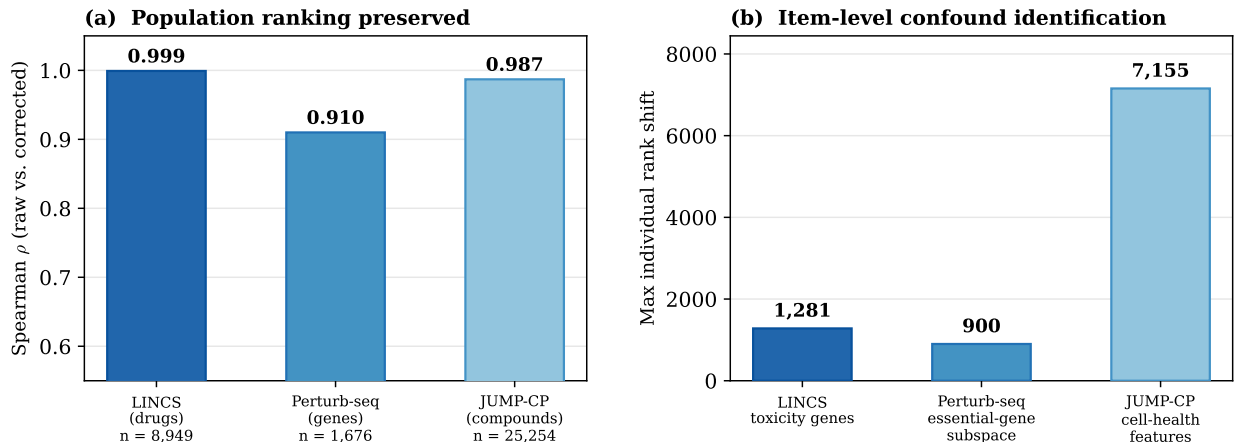


Figure 6: **Cross-domain empirical summary.** (a) Spearman ρ between raw and corrected scores across three domains. All three exceed $\rho = 0.90$: corrections preserve population rankings. Sample sizes shown below each bar. (b) Maximum individual rank shift per domain, showing that rank-preserving corrections still identify specific confound-dominated items. JUMP-CP produces the largest individual shifts (7,155 positions) from a proportionally large feature removal (23% of features), yet population $\rho = 0.987$.

not provide for most drug $\times$ cell $\times$ dose combinations.

Two additional boundary conditions affect the DAG’s blocking claim. Near-zero-norm signatures amplify noise under unit-normalization (guarded in code at $\|\mathbf{s}\| < 10^{-10}$). The Fréchet variance—the basis for the transport-stable correction (Eq. 2)—is part of the DAG’s structure rather than a component of D itself. We have not proven identification with do-calculus; the DAG is a heuristic justification for cosine normalization as a confound-blocking adjustment, not a formal identification theorem.

6.3 Cross-domain consistency

The same geometric object—angular deviation of perturbation-response directions across contexts—captures mechanism conservation in three unrelated domains (Table 5). All three exhibit the same pattern: the relevant validity condition preserves the global ranking while identifying individual perturbations whose scores are most confound-dominated.

The domain-specific failure modes differ: in drugs, cytotoxicity produces high-instability signatures; in genetics, universal-essential knockdowns produce high-distance subspaces (divergent lethal programs); in morphology, cell-health features contribute a small fraction of cross-source variance. All three invert the naive prediction that “violence mimics transport.”

6.4 Generalizability of the construct-validity procedure

The validity ladder was tested on one geometric perturbation score (direction instability) across three biological domains. The *procedure*—define interpretive threats, specify corrections that target each threat, pre-register predictions about whether corrections reorganize rankings, test empirically—applies to other geometric scores that measure perturbation-response consistency, such as connectivity scores [2], morphological similarity indices, and Grassmannian distance metrics. What would change are the domain-specific rungs: a connectivity score would require different

Table 5: Cross-domain summary. All three domains exceed the “doesn’t bite” threshold ($\rho \geq 0.83$). The LINCS HDAC gradient ρ measures fine-grained ordering within a single drug class (6 compounds in the selectivity gradient, Figure 1); the global 8,949-drug $\rho > 0.99$. H5 (transport stability) tests predictive validity rather than rank preservation; the 66-fold leave-one-out result (Table 3) provides the strongest domain-specific evidence for LINCS.

Domain	Correction	N	ρ
LINCS (HDAC gradient)	Toxicity genes	6	0.83
LINCS (global)	Toxicity genes	8,949	>0.99
Perturb-seq	Essential-gene subspace	1,676	0.91
JUMP-CP	Cell-health features	25,254	0.99

confound corrections than direction instability, but the same structure of threat identification, correction, and pre-registered testing applies. The effect size for phenotype projection ($\rho = 0.38$, explaining $\sim 14\%$ of variance) is moderate. For comparison, connectivity-score benchmarks in LINCS typically report Spearman correlations of $\rho = 0.2\text{--}0.4$ between metric and biological outcome [18], placing the phenotype-projection result at the upper end of the range achievable with unsupervised geometric scores on transcriptomic data.

6.5 Scope and limitations

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. The validity ladder is a construct-validation framework for interpretive warrant, not a surrogate endpoint. The consequential validity facet (whether warrant upgrades change practitioner decisions) is illustrated via the daunorubicin/MG-132 case study but is not empirically tested; a user study or decision-simulation experiment would be needed to validate this facet.

The drug experiments H3–H5 depend on shRNA knockdown signatures as ground-truth on-target directions. shRNA has well-documented off-target effects, which add noise to the ground truth. A CRISPRi convergent-validity check did not independently confirm the phenotype-projection result ($\rho = -0.12$, $p = 0.16$, $n = 131$); the matched 114-drug comparison (shRNA $\rho = 0.42$ vs. CRISPRi $\rho = -0.14$) points to the pre-registered K562-only confound (CRISPRi signatures from a single cell type are poor on-target directions for drugs acting in non-hematopoietic contexts). The shRNA H3 result stands on its original evidence without independent reinforcement. Drugs with unannotated targets (7,398 of 8,949) are excluded from H3 and H4, limiting these rungs to 17% of the dataset. LINCS measures 978 landmark genes, restricting the toxicity gene set to the landmark panel.

The Perturb-seq analysis uses $K = 2$ cell types, providing no Fréchet variance, holonomy, or transport stability test. Both screens target DepMap-essential genes, so the HP3 “non-essential” group consists of sub-threshold essentials rather than genuinely non-essential genes. A genome-wide screen with genuine non-essential controls would be sharper, though the label-permutation null rules out the library-overlap confound.

The transport-stable bracket formula (Eq. 2) is a heuristic linear combination of direction instability and Fréchet variance, not a principled geometric operation on the signature manifold. The penalty weight $\lambda = 1.0$ was inherited from the companion paper [3] and was not optimized for this evaluation. A pre-registered sensitivity analysis confirmed that all eight non-zero λ values

tested (0.25–5.0) pass the H5 criterion, so the result is not specific to $\lambda = 1.0$ (see §5.3). A ratio or a geodesic-based combination might perform differently. The simplicity of the linear combination is intentional: it admits straightforward interpretation of the penalty weight and enables the lambda sensitivity analysis reported in §5.3.

A pre-registered dose-direction stability test (6,169 drug×cell-line pairs) produced low within-pair cosines (median 0.19; 86.5% below 0.5), failing both pre-registered criteria. Post-hoc analysis revealed this failure is dominated by signature-level noise: 80% of dose-level groups contain a single replicate, and near-identical doses (ratio < 1.01) yield median cosine = 0.11, establishing a noise floor close to the overall result. The test lacks statistical power to distinguish genuine dose-dependent direction changes from single-replicate noise in 978 dimensions. No a priori power analysis preceded the pre-registration; replicate depth should have been assessed before committing to specific thresholds. Direction instability operates at a fixed dose across cell lines, a regime where the relevant separability condition—that cell-line response magnitude does not affect direction—is weaker than dose-independence and holds trivially. The stronger claim (direction independent of dose) remains empirically unresolved.

LINCS L1000 profiles have known batch effects across plates and production sites. Batch-driven systematic direction differences would inflate direction instability for drugs profiled across batches, producing a confound not addressed by any validity condition tested here.

Deviation entry 1 (Table 2) documents that the original pre-registration freeze omitted gene-set construction parameters; these were specified in the extended pre-registration. The gap between freezes created a potential researcher degree of freedom in gene-set construction, partially mitigated by the empirical construction (shRNA consensus from the same platform requires no pathway-database selection).

The Perturb-seq subspace dimension ($k = 5$) was fixed; geodesic distances depend on this choice, and a principled selection criterion (such as an explained-variance threshold) would strengthen the analysis.

The validity ladder draws an analogy to Bradford Hill criteria for causal inference [9]. These criteria are individually neither necessary nor sufficient for establishing causation; they are heuristic guidelines. The validity ladder shares this character: it is a structured set of proof obligations rather than a formal deductive system. Each rung eliminates a specific threat, but satisfying all tested rungs does not *guarantee* that a mechanistic claim is correct—it guarantees that specific alternatives have been ruled out.

JUMP-CP cell-health features were defined by substring matching rather than by correlation with an independent viability assay. The 78% attrition from the full dataset biases the retained set toward well-replicated compounds.

7 Conclusions

We have proposed and tested a construct-validity framework for geometric perturbation scores, using direction instability as a case study across three biological domains (drug transcriptomics, genetic perturbation subspaces, morphological profiles). Validity conditions do not improve the score’s ability to rank perturbations—the ranking is already biologically meaningful in all three domains ($\rho > 0.99$ for LINCS, 0.91 for Perturb-seq, 0.99 for JUMP-CP after correction). They improve the practitioner’s ability to make claims about that ranking and to make decisions based on those claims.

The procedure tested here—threat identification, correction specification, pre-registered prediction, empirical test—is applicable to any geometric score measuring perturbation-response con-

sistency across contexts. Phenotype projection separates on-target from off-target consistency ($\rho = 0.38$ vs. -0.04); variance correction outpredicts the raw score in all 66 leave-one-out folds ($p_{\text{Wilcoxon}} = 8.2 \times 10^{-13}$, mean $\Delta\rho = +1.10$); localization helps for intracellular enzyme targets but not receptor-mediated drugs, constraining where specific rungs apply. The domain-specific failure modes differ, but the pattern is the same: correction preserves population rankings while identifying individual perturbations most susceptible to confounding. The validity ladder provides a structured procedure for accumulating evidence that a geometric perturbation score licenses mechanistic inference.

Declarations

Ethics approval and consent to participate

Not applicable. This study uses only publicly available datasets (LINCS L1000, Perturb-seq, JUMP-CP, DepMap) with no human subjects research.

Consent for publication

Not applicable.

Availability of data and materials

All data used in this study are publicly available: LINCS L1000 (GEO GSE92742), Perturb-seq [11], JUMP-CP [12], DepMap [27]. Code, pre-registration commits (SHA 8d74bd0 for initial freeze; SHA f3c88ed, tag `prereg-extended-psb`, for extended pre-registration; SHA 5bb6080 for dose-direction separability test), deviation log, and all experimental scripts are available at <https://github.com/elliotttower/direction-instability-drug-validity>. The repository was public before the pre-registration freeze.

Competing interests

The author declares no competing interests.

Funding

This work received no external funding.

Authors' contributions

ET conceived the study, designed the validity framework, conducted all experiments, and wrote the manuscript.

Use of AI-assisted technologies

Claude (Anthropic) was used as a writing and analysis assistant during manuscript preparation. The author directed all analytical decisions and verified all numerical results against source data. The validity framework, pre-registered hypotheses, experimental results, and interpretations are the author's own. The author takes full responsibility for the content. No AI tool is listed as an author.

Acknowledgements

Not applicable.

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ET is an independent researcher focused on construct validity and geometric methods for perturbation biology.

List of abbreviations

AUROC: area under the receiver operating characteristic curve; CRISPRi: CRISPR interference; DI: direction instability; GO: Gene Ontology; HDAC: histone deacetylase; JUMP-CP: Joint Undertaking for Morphological Profiling – Cell Painting; LINCS: Library of Integrated Network-Based Cellular Signatures; LOO: leave-one-out; MOA: mechanism of action; PCA: principal component analysis; shRNA: short hairpin RNA; TS: transport-stable bracket.

Additional files

Additional file 1: Supplementary experiments and extended results. Full methods, tables, and discussion for Experiments 1, 2, and 4 (H1 toxicity correction, H2 broad-mechanism drugs, H4 localization score) and HP2 (GO pathway prediction from corrected Perturb-seq distances). Includes the complete HDAC drug list used in the sensitivity analysis. (Format: PDF)

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